Raghav Kachroo

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Software engineer drawn to making ML systems fast in production - coming from distributed systems and data pipelines, working down the stack toward inference optimization, compilers, and GPUs.

Experience

Hao AI Lab, UC San Diego | Jan 2026 – Present
Contributor | San Diego, CA

FastVideo — open-source video-generation inference and post-training framework; 4k+ stars, supported as an NVIDIA Dynamo custom worker and forked into SGLang's multimodal-generation stack. Link to PRs

- Added opt-in `cache-dit` step caching to Wan DiT, cutting wall time **18-32%** at SSIM **0.94-0.96** on Wan2.1-1.3B.
- Registered `torch.library` custom ops with `register_autograd` across FA2's default and masked/varlen paths, making attention traceable under `torch.compile` for inference and training.
- Quantized decoded frames to uint8 on-device before the GPU-to-CPU transfer, cutting end-to-end video-inference latency **15.6%** on Cosmos 2.5 2B/A100 at MS-SSIM 1.0 across all frames.
- Built an A/B harness and documentation for `torch.compile` path, surfacing **23.7%** lower latency on Wan2.1-1.3B.
- Validated FastVideo on NVIDIA DGX Spark and upstreamed a bf16 VAE-decode path at **1.2-1.3x** decode.

Amazon | Jun 2025 – Sep 2025
Software Development Engineer Intern | Bellevue, WA

- Built a distributed log indexing and query service over **42M+** log entries/hour, executing binary-search reads concurrently across a worker pool to reduce incident triage latency from **15+ minutes to under 45 seconds**.
- Automated on-call SOPs using AWS Step Functions and Lambda, saving **12+ engineer-hours per week**.
- Exposed the log query service via MCP with caching and query batching, maintaining **sub-2s** observed response times during concurrent incident investigations.

Aark Global | Apr 2023 – Sep 2024
Software Developer, AI/ML | Delhi, India

- Built an async document ingestion pipeline processing **18,000+ pages/day** via Azure Queue Storage worker pool, maintaining throughput under variable load.
- Implemented a phased Cosmos DB-to-MongoDB migration using dual writes and historical backfill while serving reads from Cosmos DB, maintaining **sub-100ms P95** latency through cutover.
- Built a full-text search pipeline ingesting scanned PDFs through OCR into indexed Elasticsearch documents, enabling **sub-180ms** query latency over previously unsearchable content.

Education

University of California, San Diego | Sep 2024 – Mar 2026
Master of Science in Data Science

Indraprastha Institute of Information Technology, Delhi | Sep 2022 – Sep 2023
Post Graduate Diploma in Data Science & Artificial Intelligence

University of Delhi | Jul 2018 – Jun 2021
Bachelor of Management Studies

Projects & Research

Flare: ML-Powered Anomaly Detection Pipeline | *GitHub* | Jan 2026 – June 2026

- Built an end-to-end ML anomaly detection pipeline applying Drain3 template mining, Isolation Forest scoring, and DBSCAN clustering to **11.2M log lines** across **575K blocks**, achieving **0.642 F1** on a held-out split at ∼**40ms** scoring latency per block (CPU-only).

OneBusAway / maglev: OSS Contributions | *GitHub* | Jul 2026 – Present

- Landed **9** upstream fixes to this Go transit REST API from **9** spec-conformance issues I filed - query radius, sort order, `maxCount` clamping, route-type filtering, and `includeReferences` handling.

TRAI: AI Mobile App to Reduce the Gap Between Triage and Care | *IEEE Publication* | Jul 2025

- Co-authored a IEEE paper on an LLM-based mobile application for patient triage and clinical data gathering.

Technical Skills

Languages: Python, Go, SQL, Java
ML Systems: PyTorch, torch.library, FlashAttention, Nsight Systems, Diffusers, Quantization
Backend & Infra: FastAPI, REST APIs, Docker, Kubernetes, CI/CD, GitHub Actions, Linux, MCP
Distributed & Data Systems: Kafka, Spark, Airflow, Elasticsearch, MongoDB, Cosmos DB, Azure Queue Storage, Redis
Observability & Cloud: Prometheus, OpenTelemetry, AWS (Lambda, Step Functions, S3), Azure
Concepts: Distributed Systems, Concurrency, Inference Optimization, MLOps, System Design, RAG